

Global Health & Life Expectancy Dashboard

**AnalystLab Africa — Batch B Internship Final
Capstone Project**

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Project Overview

This project analyzes global health and life expectancy trends using the World Bank's **World Development Indicators (WDI)** dataset. It examines how life expectancy has evolved since 1960, how it varies by region and income group, and how it relates to health expenditure and maternal mortality — culminating in an interactive Power BI dashboard.

Objective

To identify which regions and income groups lag behind in health outcomes, and to assess whether health expenditure (as % of GDP) reliably predicts better outcomes — supporting evidence-based recommendations for global health stakeholders.

Dataset

- **Source:** [World Bank World Development Indicators](#)
- **Files used:** `WDICSV.csv` (main indicator data), `WDICountry.csv` (region/income group metadata)
- **Coverage:** 217 countries/territories + regional/income aggregates (264 entities), 1960–2024
- **Key indicators:** Life expectancy at birth, current health expenditure (% of GDP), maternal mortality ratio

Tools Used

- **Power Query (Power BI):** data cleaning, filtering, unpivoting, and merging
- **Power BI Desktop:** dashboard design and interactive visualization
- **Microsoft Word / PDF:** final report

Data Cleaning Steps

1. Promoted headers and set correct data types
2. Filtered to health-related indicators only
3. Unpivoted year columns (wide → long format)

4. Removed nulls and duplicate rows
5. Merged in Region and Income Group from country metadata

Full details are documented in the **Final Report (PDF)**.

Dashboard Features

- **KPI Cards:** Avg. Life Expectancy (2022), Avg. Health Expenditure (% GDP), Maternal Mortality Ratio, Countries Covered
- **Trend Chart:** Life expectancy by region, 1960–2024
- **Comparison Charts:** Top 10 countries by life expectancy; Top 15 countries by health expenditure (% GDP)
- **Slicers:** Region, Income Group, Year — for interactive filtering

Key Findings

- A persistent life expectancy gap remains between Sub-Saharan Africa (~65 years) and other regions (~75–83 years)
- Top life-expectancy countries are mostly small, high-income territories (Monaco, San Marino, Liechtenstein) rather than large economies

- High health expenditure as % of GDP does not reliably predict top life-expectancy rankings
- Global maternal mortality remains high at ~133 deaths per 100,000 live births
- A visible dip in life expectancy appears across most regions around 2020–2021, consistent with COVID-19 impact

Full insights and recommendations are in the **Final Report (PDF)**.

Repository Contents

```
|—  
Final_Report_Health_Life_Expectancy.pdf  #  
Full written report  
|— Global_Health_Dashboard.pbix  
# Power BI dashboard file  
|— screenshots/  
# Dashboard screenshots  
|— README.md  
# This file
```

How to View

1. Open the `.pbix` file in **Power BI Desktop** (free download from Microsoft) to explore the interactive dashboard

2. Or view the static **screenshots** and **PDF report** for a non-interactive summary

Demo Video

[Insert your demo video link here — Google Drive / YouTube]

Acknowledgment

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